Garrett Comes

+1 (404) 981-3420 | garrett.comes@gatech.edu | garrettcomes24@gmail.com
linkedin.com/in/garrett-comes | github.com/GHawk1124 | garrettcomes.me

Summary

Aerospace engineer trained at Georgia Tech with a propulsion concentration, equally at home in spacecraft fluid hardware and the software that analyzes it. Four summers at Lockheed Martin Space on flight programs including Dragonfly and Mars Sample Return. Builds simulation tools in Rust, Python, and CUDA: GPU CFD solvers, superconducting-coil multiphysics, and orbital analysis, plus independent research on thermodynamic computing hardware. Strong background in engineering optimization and agentic AI tooling.

Education

Georgia Institute of Technology **Jul 2022 – Aug 2026**
B.S. Aerospace Engineering, Concentration in Propulsion *Atlanta, GA*

- Coursework: Jet & Rocket Propulsion, Computational Fluid Dynamics, Aerothermodynamics, Finite Element Analysis, Control System Design, Aircraft Flight Dynamics, Aerodynamics.

Professional Experience

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space **Summers 2022 – 2025**
Flight programs incl. Dragonfly and Mars Sample Return *Waterton Campus, Denver, CO*

- Designed a piston-tank propellant gauging system from a blank sheet: internal hermetically sealed magnets sensed by radiation-hardened Hall-effect sensors; owned magnetic circuit sizing, sensor selection, magnetic simulation, and integration concept, then correlated analysis with bench testing.
- Built propellant slosh CFD simulations of spacecraft tanks in Ansys and supported design of a slosh characterization test rig used to correlate model predictions against test data.
- Contributed to a green-propellant trade study evaluating hydrazine-replacement monopropellants across performance, materials compatibility, and handling/safety criteria.
- Designed and analyzed propulsion hardware and fluid-system components — valves, interfaces, and test support equipment — including Creo CAD models, layouts, and drawings for propulsion assemblies.
- Supported propulsion hardware test campaigns, requirements verification, anomaly investigation, and cross-functional design reviews with systems, manufacturing, and test teams.

Boys & Girls Clubs of the Valley — Future Engineers Intern **Jul – Aug 2021**
Onsite engineering and construction, Valley Metro Northwest light-rail expansion *Phoenix, AZ*

Publications

- **SCRAP: Contactless Detumbling and Capture of Orbital Debris via HTS Coil** — extended abstract in preparation for AIAA SciTech 2027, 2026; co-authors from Lockheed Martin, Astranis, and Boeing. [draft]

Independent Research

Thermodynamic Sampling Unit Research | *JAX, THRML, PyTorch, flash-attn* **2026**

- Hybridized a pretrained masked-diffusion language model with block-Gibbs joint sampling over a factor graph on Extropic's THRML thermodynamic-hardware simulator; joint factor-graph decoding beat independent argmax by 7–15 pp on a held-out test split, with hard-equality factor maps dominating.
- Ran fail-fast probes on the "no classical escape" hypothesis and retired it: warm-started AIS/SMC crossed every tested energy barrier, closing the per-sample-accuracy axis for TSU advantage claims.
- Quantified a sampling coverage frontier across 59 iid $q=20$ Potts landscapes of the protein-design coevolutionary type; derived the end-to-end hardware energy ratio required for crossover at digital budgets and labeled it explicitly as unmeasured.
- Subjected all headline claims to adversarial review gates that scoped an initial $10^6\times$ energy claim into a precise conditional one; the final pipeline reproduces bit-for-bit with 9/9 regression tests passing.

Engineering Projects

SCRAP — Orbital Debris Detumbling & Capture, Senior Design | *Python, FEniCSx, Rust, Gmsh* **2025 – 2026**

- Developed a coupled electromagnetic–thermal quench simulation of the YBCO HTS pancake coil, using an H-formulation with E–J power-law resistivity, per-turn tape meshing, radiation and cold-finger BCs, quench detection, and adaptive time-stepping, to establish safe operating current margins.
- Trained DDPG reinforcement-learning agents for eddy-current detumbling of uncooperative tumbling targets.

spacedb — Space-Object Database & Servicing Route Planner | *Rust, egui, wgpu* **2025 – 2026**

- Desktop GUI + CLI over the full public space-object catalog from Space-Track, GCAT, and ESA DISCOS: SGP4 and high-fidelity propagation, collision screening, decay prediction, debris value scoring, and multi-target servicing-route optimization; wgpu-rendered interactive 3D Earth/orbit scene.

Eddy-Current Braking System — Controls & Hardware | *Rust no_std, Arduino Due, NGSolve* **2025**

- Built an eddy-current brake end to end: no_std Rust firmware on an Arduino Due reading an AS5048A magnetic encoder over SPI and driving an H-bridge with PWM, with explicit control modes for manual duty, feedforward, and feedforward-plus-PI speed control, output clamps, slew limits, and target ramping.
- Paired the firmware with a live egui tuning dashboard over UART telemetry, plotting motion and internal controller signals in real time, plus Python/NGSolve magnetic simulations of the braking physics; the whole stack, from flashing to simulation, runs under Nix.

Valurile — GPU Lattice-Boltzmann CFD Solver | *Rust, CUDA* **2024 – present**

- Multi-crate LBM workspace: Esoteric Pull in-place streaming that roughly halves memory traffic per timestep, static mesh refinement, Smagorinsky–Lilly LES, SRT/TRT/MRT collision operators, airfoil/cylinder obstacles with drag/lift histories, and interactive egui/GPU visualization.

fem_rust_ae — FEM Solver in Rust, live in the browser | *Rust, WASM, Bevy* **2025**

- From-scratch structural FEM, a shell model of a cylinder under axial load, with sparse solves via faer, an interactive Bevy/egui 3D UI, mesh-refinement convergence studies, and a WebAssembly build running live in the browser.

ai-cfd — GPU-Resident Flow Solver for ML Dataset Generation | *Python, Triton* **2026**

- Staggered-grid incompressible cylinder-flow solver written entirely in Triton kernels, from momentum through projection to rendering; adaptive timestep, Jacobi/CG/BiCGStab pressure options, per-step physics diagnostics, and train/val/test sweep generation for AI-CFD surrogate experiments.

repx — eBPF Reproducible-Build Attestation | *Rust, Aya eBPF* **2026**

- Kernel-level tracing of build process/file events canonicalized into Merkle-rooted attestations with content-addressed dataflow-DAG provenance; verification reruns report divergent tree nodes, not just pass/fail.

any-nix-bootstrap — Trust-Minimized Software Bootstrap | *POSIX sh, live-bootstrap, Guix, Nix* **2026**

- Bootstrap chain from an 813-byte auditable seed through GCC 15/musl/Guile, GNU Guix, and Nix to any nixpkgs package, fully offline from one hash-verified source manifest; stage 1 executed end-to-end on real hardware against 314 hash-pinned distfiles.

Stream Sim — Browser Fluid Simulation | *JavaScript, Canvas, Pyodide* **2023**

- Grid-based incompressible-flow demo running live in the browser: staggered velocity fields, pressure projection, semi-Lagrangian advection, airfoil obstacles, and streamline/smoke visualization with interactive controls.

Nix Packaging for Claude Desktop & Claude Science | *Nix flakes, NixOS* **2025 – 2026**

- Published claude-desktop-nix and claude-science-nix, flakes packaging Anthropic's Linux apps for NixOS, incl. a bubblewrap overlay so the apps' own sandboxing resolves bash, the dynamic linker, and CA certificates.

Skills

Languages: Python, Rust, C++, MATLAB/Simulink, CUDA/Triton, Wolfram, LaTeX, shell
Simulation & CAE: Ansys Fluent and Mechanical, FEniCSx, NGSolve, Gmsh, SolidWorks, Creo, Fusion 360, Inventor
Scientific computing: JAX, PyTorch, NumPy/SciPy, wgpu/OpenGL, MPI/PETSc, probabilistic inference incl. Gibbs, AIS, and SMC, engineering optimization
Aerospace: space propulsion & fluid systems, valves, propellant management, orbital mechanics incl. SGP4 and high-fidelity propagation, CFD & slosh analysis, magnetic simulation & test
Systems & tooling: Linux/NixOS, Git, eBPF, embedded no_std Rust, agentic AI development with Claude Code, MCP servers, and custom agent frameworks

Leadership, Certifications & Awards

Yellow Jacket Space Program **2023 – present**
Propulsion — pressurant isolation valve design for a collegiate liquid-propellant space-shot vehicle *Georgia Tech*

Apex Robotics, FRC and VEX **2017 – 2022**
President & Lead Software Engineer — competition robots, outreach, fundraising, team coordination *Mesa, AZ*

Stellar Xplorers **2021 – 2022**
Team Captain — led team to one of ten national-finals spots in Houston *Mesa, AZ*

Certifications: SolidWorks CSWA · Autodesk Inventor · Autodesk Fusion 360 · Wolfram Language Level 1 · **Awards:** CTE Presidential Scholar, National Semifinalist, 2022